



Research paper

Coupled and decoupled aero-hydro-geotech reliability analyses of monopile-supported offshore wind turbines considering multi-source uncertainty propagation

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ABSTRACT

The safety of monopile-supported offshore wind turbines (OWTs) is subjected to multi-source uncertainties. The industrial split-design practice decouples the analyses of upper structure and foundation, which neglects complex aero-hydro-geotech feedback and often underestimates geotechnical uncertainties. This study establishes a fully coupled aero-hydro-geotech reliability analysis framework to quantify how multi-source uncertainties affect the safety of OWTs. It integrates soil-pile interaction modeling, time-domain dynamic simulation, and uncertainty modeling into the OpenFAST platform. A simplified decoupled method is also developed to benchmark the industrial split-design practice. These methods are validated and compared through an NREL 5 MW reference OWT under three load cases. Geotechnical uncertainties can be properly considered through a variable pile-head stiffness matrix in soil-pile interaction modeling and ignoring geotechnical uncertainties will result in a modest overestimation of reliability. More importantly, a counter-intuitive “false sense of security” emerges from the decoupled method. Although it yields higher mean responses due to the simplification of rigid foundation, the truncation of uncertainty propagation significantly reduces response variabilities, resulting in non-conservatively high reliability indices, with an overestimation ranging from 15% to 60%. The coupled method can provide a robust tool for design code calibration under multi-source uncertainty propagation to mitigate the excessive design redundancy issue.

1. Introduction

Offshore wind energy has emerged as a pivotal component of the global renewable energy strategy, distinguished by its superior, persistent wind resources and vast availability of space (Arshad and O’Kelly, 2013). The monopile foundation remains the prevailing support structure for offshore wind turbines (OWTs), particularly in shallow waters with depths less than 30 m (Qin et al., 2020). The anticipated expansion in turbine scale, driven by improvements in power generation and wind capture efficiency, has enhanced overall energy capture capabilities and reduced the levelized cost of energy (e.g., Musial et al., 2023; Nia and Edalat, 2025). In this context, balancing cost-efficiency with system reliability, specifically, avoiding unnecessary expenditures while ensuring structural safety, has become a critical challenge (Morató and Sriramula, 2021). Consequently, recent research has increasingly shifted toward probabilistic design optimization of OWTs (e.g., Okpokparoro

and Sriramula, 2021; Ramezani et al., 2023; Shin et al., 2023; Li et al., 2024; Lin et al., 2024a). It replaces the traditional deterministic safety factor with a reliability index to better represent the structural performance of OWTs, through parametric uncertainty characterization and reliability analysis. Among various reliability methods, surrogate models (e.g., Meng et al., 2023; Meng et al., 2024; Quevedo-Reina et al., 2025; Zhang and Noshadravan, 2025) are widely used for efficient reliability evaluation of OWTs.

During their service life, OWTs are subjected to complex, multi-source uncertainties, necessitating rigorous probabilistic safety assessments to prevent catastrophic economic losses. From a system engineering perspective, the primary drivers of uncertainty can be categorized into metocean conditions (i.e., aerodynamic and hydrodynamic loads) and subsurface geotechnical conditions. While the stochastic nature of metocean loads has been extensively characterized in existing literature, the inherent spatial variability and parametric

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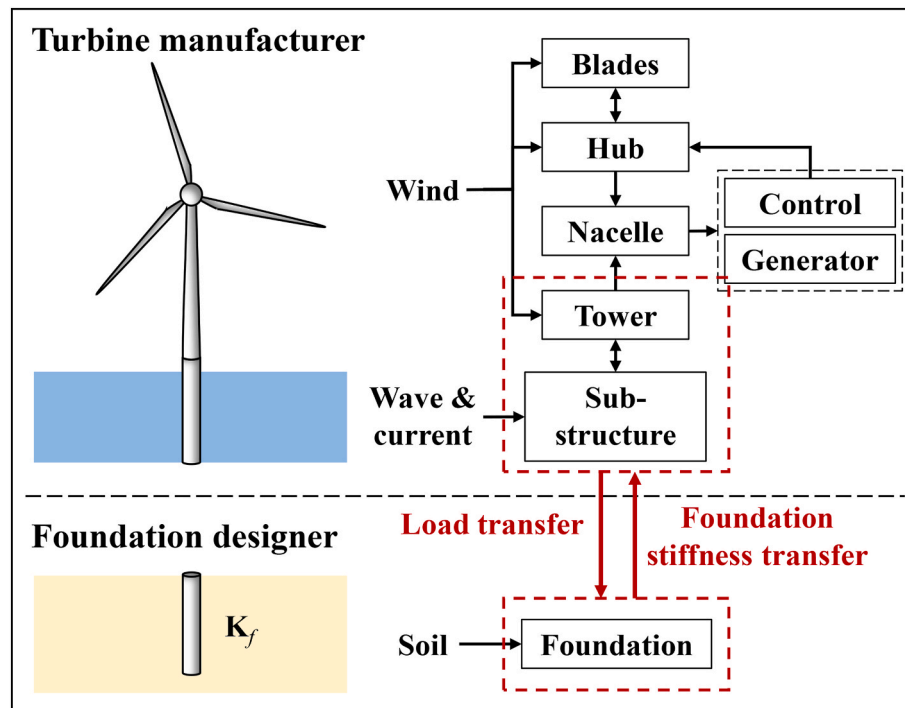


Fig. 1. Traditional decoupled design workflow for monopile-supported OWTs.

uncertainty of seabed soils remain largely underexplored in OWT studies (e.g., Xiao et al., 2021; Zhang et al., 2023; Wang et al., 2025; Han et al., 2025; Zhang et al., 2025). This discrepancy is largely attributed to the high economic cost associated with deep offshore site investigation and the complexity of soil spatial variability modeling with limited data. Nevertheless, dynamic soil-pile interaction is a critical determinant of the OWT system response, particularly for low-redundancy monopile foundations. Although soil-pile interaction has been extensively investigated (e.g., Arany et al., 2016; Krathe and Kaynia, 2017; Yang et al., 2019; Alkhoury et al., 2022; Wang and Ishihara, 2022; Han et al., 2024; Zhang et al., 2026), most studies focus on refining deterministic mechanical models, such as integrating nonlinear hysteretic springs or employing high-fidelity finite element models (e.g., Bouzid et al., 2018), rather than quantifying the impact of soil-pile interaction uncertainties on the reliability of OWT system.

An OWT represents a complex coupled dynamic system, theoretically necessitating time-domain analysis using fully integrated multibody simulation tools like OpenFAST or HAWC2. However, in industrial practice, implementing such fully coupled probabilistic modeling faces significant technical challenges. Beyond computational costs, a primary barrier is the strict division of design responsibilities and intellectual property protection between turbine manufacturers and foundation designers, which restricts the access to detailed upper structure data for foundation design (van der Male et al., 2020). Consequently, rather than employing a fully integrated framework, the industry typically relies on decoupled methods to simplify the design process, as shown in Fig. 1, in which upper structure and foundation are designed separately. Wind and wave loads acting on the blades, tower, and substructure are often simplified as equivalent resultant forces and applied to an independent foundation model accounting for soil resistance. Mutual iteration of the two parts is necessary, but this practice is seldom adopted in reality.

In this aspect, Chen et al. (2020) and Yu et al. (2023) incorporated wind loads into ABAQUS to analyze monopile dynamics under combined wind and wave loading. Similarly, Morató et al. (2019) simplified the wind, structural dynamics, and hydrodynamic loads acting on the rotor and tower into resultant forces at the tower top, which were then applied to a finite element model of the pile. However, this physical

decomposition severs the bidirectional dynamic feedback loop (e.g., the beneficial suppression of hydrodynamic responses by aerodynamic damping) and artificially truncates the continuous propagation of multi-source uncertainties from the environment down to the soil. To compensate for this loss of physical interaction and uncertainty transmission, engineers are forced to apply highly conservative load factors at the design interface. While intended to ensure safety, this deterministic compensation often leads to excessive design redundancy, resulting in unnecessarily oversized foundations and inflated project costs (Passon and Branner, 2014; Smilden, 2019). There remains a lack of research quantifying the error introduced by this truncated uncertainty propagation in split-design approaches compared to a fully coupled baseline.

In light of the above, this study aims to address two critical issues: (1) the systematic underestimation of geotechnical uncertainties in current probabilistic frameworks, where deterministic foundation models fail to reflect the significant variability of actual geological conditions, despite its vital role in monopile response; and (2) the inherent limitation of industry-standard decoupled design methods, which artificially truncate the propagation of multi-source uncertainties without bidirectional dynamic feedback, alongside a lack of quantitative error assessment between fully coupled and simplified decoupled models.

This study establishes a fully coupled OWT analysis framework accounting for multi-source uncertainties (aerodynamic, hydrodynamic, and geotechnical) and quantifies the contribution of different uncertainties to key structural responses. A simplified decoupled model featuring equivalent interface load transfer but lacking aero-hydro-geotech feedback is proposed to mimic the industrial split-design process. This allows for a comparative evaluation of structural reliability using the decoupled and coupled approaches. The remainder of this paper is organized as follows: Section 2 presents the coupled reliability analysis framework for OWTs under multi-source uncertainties and defines the limit state functions, followed by the development of a decoupled framework in Section 3. Section 4 presents a case study using both coupled and decoupled methods. Finally, Section 5 summarizes the main results.

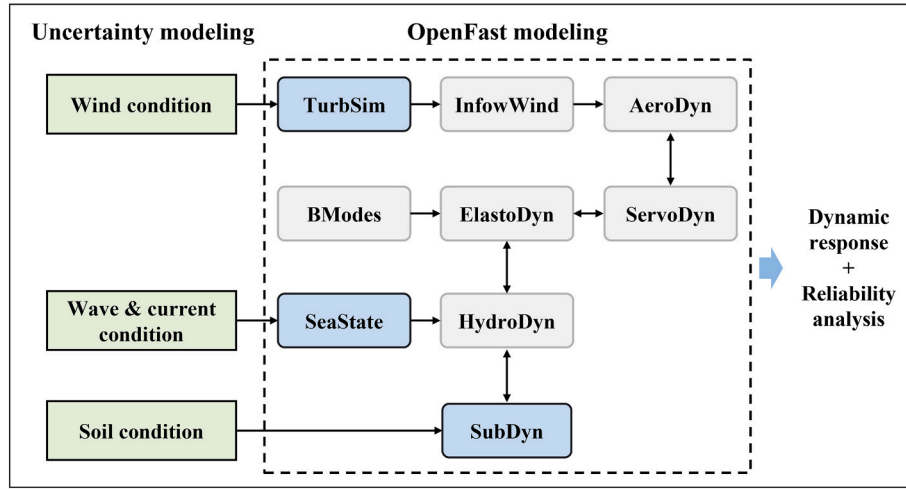


Fig. 2. Framework for coupled reliability analysis of OWTs under multi-source uncertainties.

2. Coupled reliability analysis framework

2.1. Coupled dynamic modeling strategy

2.1.1. Modeling with OpenFAST

In this study, the fully coupled dynamic analysis of OWT is performed using OpenFAST (Jonkman and Shaler, 2021), a comprehensive aero-hydro-servo-elastic simulation tool designed for nonlinear time-domain analysis. OpenFAST integrates modules governing aerodynamics, hydrodynamics for offshore structures, control and electrical system dynamics, and structural dynamics to simulate the response of wind turbines. Evolving from FAST v8, OpenFAST has undergone continuous updates and introduced substantial improvements, such as the SeaState module for wave field definition. The latest version (v4.1.2, released in August 2025) allows the SubDyn module to operate independently without invoking HydroDyn module when applying external loads. This capability is pivotal for the present study, as it serves as the foundation for the simplified decoupled method proposed in Section 3.

Fig. 2 illustrates the interaction between different modules in OpenFAST and the uncertainty modeling workflow for aerodynamic and hydrodynamic loads and geotechnical resistances in reliability analysis. It should be highlighted that the SubDyn module is a bridge connecting the upper structure and the foundation. The module utilizes the Craig-Bampton reduction method (Jonkman and Shaler, 2021) to conduct finite element analysis of the OWT foundation system with the consideration of soil-pile interaction.

2.1.2. Soil-pile interaction modeling

In the OC3 (Offshore Code Comparison Collaboration) project (Jonkman and Musial, 2010), it is found that the bearing soils of a monopile usually possess sufficient vertical stiffness to eliminate notable vertical deformation. Hence, most efforts focus on modeling the lateral resistance of a monopile. Three primary soil-pile interaction models have been developed for this purpose, i.e., the apparent fixity model, the coupled springs model, and the distributed springs model. The coupled springs model characterizes the soil-pile interaction by introducing a localized stiffness and damping matrix at the mudline and can capture the six rigid-body degrees of freedom of the monopile. While simplifying the sub-mudline structural behavior, the model preserves the global boundary conditions and the overall dynamic feedback at the seabed interface. Given the paramount need for computational efficiency when propagating multi-source uncertainties in the fully coupled dynamic analysis, this study adopts the coupled springs model for soil-pile interaction modeling.

In the context of coupled springs model, the stiffness matrix of

monopile foundation at the mudline, $\mathbf{K}_f$, consists of two parts, i.e., the original stiffness matrix of the pile ($\mathbf{K}_{pile}$) and a linearized pile-head stiffness matrix ($\mathbf{K}_{soil}$) considering soil-pile interaction. The latter can be written as:

$$\mathbf{K}_{soil} = \begin{bmatrix} k_{xx} & 0 & 0 & 0 & -k_{x\theta_y} & 0 \\ 0 & k_{yy} & 0 & k_{y\theta_x} & 0 & 0 \\ 0 & 0 & k_{zz} & 0 & 0 & 0 \\ 0 & k_{\theta_x y} & 0 & k_{\theta_x \theta_x} & 0 & 0 \\ -k_{\theta_y x} & 0 & 0 & 0 & k_{\theta_y \theta_y} & 0 \\ 0 & 0 & 0 & 0 & 0 & k_{\theta_z \theta_z} \end{bmatrix} \quad (1)$$

where k_{xx} , k_{yy} , and k_{zz} are the translational stiffness components along x , y , and z axes, respectively; $k_{\theta_x \theta_x}$, $k_{\theta_y \theta_y}$, and $k_{\theta_z \theta_z}$ are the rotational stiffness components; and the other terms characterize the cross-interaction between translational displacement and rotation. In this study, $\mathbf{K}_{soil}$ is obtained using a numerical approach based on the Winkler foundation model (beam-on-elastic-foundation theory), in which the stiffness components can be calculated as (Liu et al., 2024):

$$k_{xx} = k_{yy} = \frac{f_{HH}}{f_{HH}f_{OM} - f_{HM}f_{OH}} \quad (2)$$

$$k_{\theta_x \theta_x} = k_{\theta_y \theta_y} = \frac{f_{OM}}{f_{HH}f_{OM} - f_{HM}f_{OH}} \quad (3)$$

$$k_{x\theta_y} = k_{y\theta_x} = \frac{f_{HM}}{f_{HH}f_{OM} - f_{HM}f_{OH}} \quad (4)$$

$$k_{zz} = a \frac{A_p E_p}{L} \quad (5)$$

$$k_{\theta_z \theta_z} = \frac{GJ}{L_0} \quad (6)$$

where f_{HH} and f_{HM} are the pile-head horizontal displacements induced by a unit horizontal load and a unit moment, respectively; f_{OH} and f_{OM} are the corresponding pile-head rotations; parameter a is related to pile type and can be taken as $a = 0.027L/D + 0.2$, in which D and L are the diameter and embedded length of the pile, respectively; L_0 is the equivalent fixity depth and it is taken as $L/4$ for simplicity in this study; A_p is the net cross-sectional area of the pile; E_p and G are the elastic modulus and shear modulus of the pile, respectively; and J is the torsional moment of inertia.

Since soil uncertainties will be considered in this study, the stiffness matrix $\mathbf{K}_{soil}$ varies in different random realizations. The stochastic $\mathbf{K}_{soil}$ is mapped into the SubDyn module first, and then used to assemble the

foundation stiffness matrix K_f to participate in the fully coupled dynamic analysis of the OWT. By this means, the OWT is supported by a flexible foundation with a robust consideration of soil-pile interaction and soil uncertainties.

2.2. Modeling of multi-source uncertainties

The probabilistic structural reliability analysis lies in the rational quantification of uncertainties. As outlined previously, uncertainties affecting the load and resistance effects on OWTs have three primary sources: aerodynamic, hydrodynamic, and geotechnical. They correspond to stochastic variations in wind, wave/current, and soil, where the latter translates directly into uncertainties in soil-pile interaction. Mathematically, these multi-source uncertainties can be represented by a vector of basic random variables, defined as:

$$\Theta = \{\Theta_{wind}, \Theta_{sea}, \Theta_{soil}\} \quad (7)$$

where Θ_{wind} , Θ_{sea} , and Θ_{soil} are the sub-vectors of random variables characterizing the aerodynamic environment, hydrodynamic conditions, and geotechnical properties of supporting soils, respectively. These variables are sampled to generate stochastic input fields and soil stiffness matrices for the fully coupled simulations described in Section 2.1. The stochastic models adopted for each source are summarized in the following subsections.

2.2.1. Random wind field

The turbulent wind field at the rotor plane can be represented as the sum of a mean component and a fluctuating component:

$$U(z, t) = U_{ave}(z) + U'(z, t) \quad (8)$$

where $U_{ave}(z)$ is the mean wind speed at height z ; and $U'(z, t)$ is the zero-mean turbulence. The vertical profile of the mean wind speed can be described by (Bhattacharya, 2019):

$$U_{ave}(z) = U_w \cdot (z/z_{hub})^r \quad (9)$$

where U_w is the mean wind speed at the hub height; z_{hub} is the hub height above sea level; and the power-law exponent $r = 0.14$ is adopted as a representative value for offshore sites. The turbulent fluctuations are generated from a prescribed one-sided Kaimal spectrum using an inverse fast Fourier transform. For each velocity component $k = \{x, y, z\}$, the spectrum $S_k(f)$ is given by (Bhattacharya, 2019):

$$S_k(f) = \frac{4\sigma_k^2 L_k / U_w}{(1 + 6fL_k / U_w)^{5/3}} \quad (10)$$

where f is the frequency; σ_k is the standard deviation; and L_k is the integral length scale of the k -th component. The quantities U_w , σ_k and L_k are determined from the target mean wind speed, turbulence intensity, and IEC turbulence class for each load case. Among all these parameters, only U_w is modeled as a random variable that controls the aerodynamic uncertainty vector Θ_{wind} .

2.2.2. Random sea state

The stochastic hydrodynamic environment originates from wave and current contributions. The current is simply characterized by the current speed U_c , whereas the wave is primarily characterized by the significant wave height (H_s), which defines the overall energy intensity of the irregular sea state, and the spectral peak period (T_p), corresponding to the frequency at which the wave energy is concentrated. In this study, the sea surface elevation is modeled as a stationary Gaussian process via the linear superposition of sinusoidal wave components, with amplitudes derived from the JONSWAP spectrum. The spectral density, $S_\eta(\omega)$, is expressed as (Hasselmann et al., 1980):

$$S_\eta(\omega) = \alpha g^2 \omega^{-5} \exp \left[-\frac{5}{4} \left(\frac{\omega_p}{\omega} \right)^4 \right] \gamma^{\exp \left[-\frac{(\omega - \omega_p)^2}{2\sigma^2 \omega_p^2} \right]} \quad (11)$$

where ω is the wave angular frequency; $\omega_p = 2\pi/T_p$ is the peak angular frequency corresponding to the peak period T_p ; g is the gravitational acceleration; γ is the peak enhancement factor; σ is chosen in accordance with typical offshore conditions and IEC 61400-3-1 recommendations; and α is the Phillips constant that is determined by H_s and T_p to ensure the total energy of the spectrum matches the target sea state as follows (Bhattacharya, 2019):

$$\alpha = \frac{5H_s^2 \omega_p^4}{16g^2} (1 - 0.287 \ln \gamma)^{-1} \quad (12)$$

Consequently, U_c , H_s and T_p are modeled as random variables that control the hydrodynamic uncertainty vector Θ_{sea} . For operational load cases, H_s and T_p are modeled according to their probability distributions calibrated to the long-term joint sea-state statistics at the site, while for extreme load cases, they are modeled around the characteristic extreme values with an appropriate variability. If necessary, extreme wave groups or breaking waves should be properly considered under extreme sea state (Zhao et al., 2024).

2.2.3. Random soil profile

The soils supporting the monopile are another important source of uncertainties. The soil profile can be discretized into several horizontal layers, each characterized by a set of random soil parameters to reflect limited site investigation data, measurement errors and inherent spatial variability (Phoon and Kulhawy, 1999). Particularly, this study characterizes the spatial variability by modeling the thickness variation of soil layers, instead of involving a spatial correlation function within each layer (e.g., Zhang et al., 2023; Li et al., 2024). By this means, taking sandy soils as an example, the uncertain soil vector Θ_{soil} can be written as:

$$\Theta_{soil} = \{\varphi_i, \gamma_i, h_i\}, i = 1, 2, \dots, n_l \quad (13)$$

where φ_i , γ_i and h_i are the internal friction angle, effective unit weight, and layer thickness of the i -th soil layer, respectively; and n_l is total number of soil layers concerned. All these random variables are generated from certain probability distributions and eventually mapped into the elements of the pile-head stiffness matrix K_{soil} used in the coupled springs model.

2.3. Limit state functions and reliability analysis

In the design of offshore foundations, the lateral displacement and rotation of the monopile at the seabed level (mudline) are the most critical performance indicators. For serviceability limit state, the DNV-ST-0126 code (Det Norske Veritas, 2021) mandates the prediction of accumulated permanent rotation at the mudline, and the IEC 61400-3-1 code (International Electrotechnical Commission, 2019) specifies an allowable permanent rotation of 0.5° at the seabed (Lin et al., 2024a). Regarding the ultimate limit state, DNV requires that the maximum pile-head deflection at the mudline remain within prescribed limits. A widely accepted criterion for the allowable lateral displacement at the seabed is $0.1D$ (Lin et al., 2024b), where D is the pile diameter.

For reliability analysis of OWTs under multi-source uncertainties, this study evaluates the failure probabilities of the monopile foundation corresponding to the displacement limit state function $G_y(\Theta)$ and the rotation limit state function $G_\theta(\Theta)$, as follows:

$$\begin{cases} G_y(\Theta) = y_{lim} - y_{cal}(\Theta) \\ G_\theta(\Theta) = \theta_{lim} - \theta_{cal}(\Theta) \end{cases} \quad (14)$$

where $y_{lim} = 0.1D$ and $\theta_{lim} = 0.5^\circ$ are allowable lateral displacement and

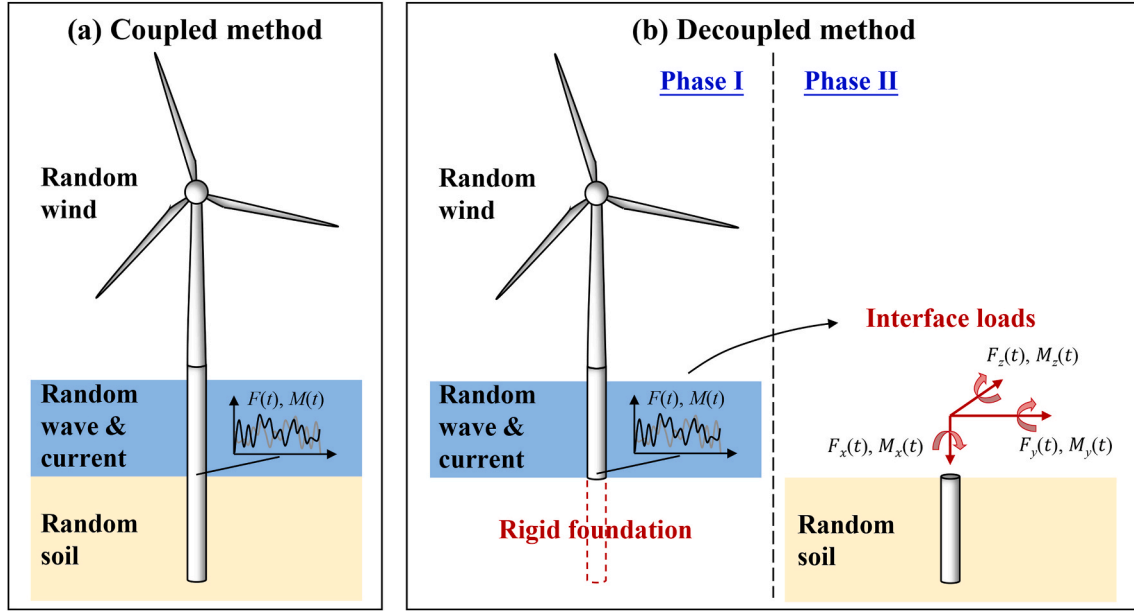


Fig. 3. OWT reliability analysis: (a) coupled method; (b) decoupled method.

Table 1
Key parameters of the NREL 5 MW reference OWT.

Parameter	Value
Rating	5 MW
Rotor orientation, configuration	Upwind, 3 blades
Control	Variable speed, collective pitch
Drivetrain	High speed, multiple-stage gearbox
Rotor, hub diameter	126 m, 3 m
Hub height	90 m
Cut-in, rated, cut-out wind speed	3 m/s, 11.4 m/s, 25 m/s
Cut-in, rated rotor speed	6.9 RPM, 12.1 RPM
Rated tip speed	80 m/s
Overhang, shaft tilt, precone	5 m, 5°, 2.5°
Rotor mass	110,000 kg
Nacelle mass	240,000 kg
Tower mass	347,500 kg
Coordinate location of overall center of mass	(-0.2 m, 0.0 m, 64.0 m)

Table 2
Selected load cases.

Load case	Case 1 (LC 5.2)	Case 2 (DLC 1.3)	Case 3 (DLC 6.2a)
Wind speed U_w	11.4 m/s (NTM)	14 m/s (ETM)	40.375 m/s (EWM)
Wave height H_s , wave period T_p	6 m, 10 s (SSS)	1.91 m, 6.07 s (NSS)	8.07 m, 11.3 s (ESS)
Current speed U_c	0.6 m/s (NCM)	0.6 m/s (NCM)	1.2 m/s (ECM)
Yaw misalignment	8°	8°	90°
Wind/wave angle	0°	0°	30°
Other condition	—	—	Loss of electrical network

*Note: NTM: normal turbulence model; ETM: extreme turbulence model; EWM: extreme wind speed model; NSS: normal sea state; SSS: severe sea state; ESS: extreme sea state; NCM: normal current model; ECM: extreme current model.

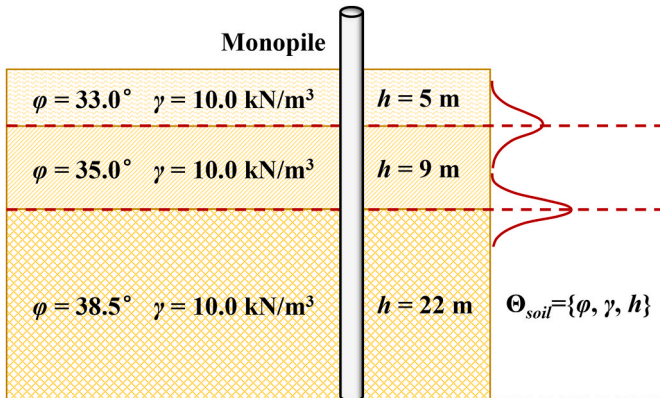


Fig. 4. Stratigraphy and geotechnical profile of the reference site.

rotation angle, respectively; and $y_{cal}(\Theta)$ and $\theta_{cal}(\Theta)$ are the computed maximum lateral displacement and rotation angle, respectively, at the mudline. In this study, the lateral displacement y_{cal} can be directly obtained from the dynamic analysis of OWTs, while the rotation angle θ_{cal} is estimated from y_{cal} using $\theta_{cal} = \sin^{-1}[y_{cal}/(b \cdot L)]$ (Lin et al., 2024a),

where L is the embedded pile length; and $b = 0.75 \sim 0.8$ is a rotation center modification factor and $b = 0.775$ is adopted. Monopile failure is defined by the criterion $G_y(\Theta) < 0$ or $G_\theta(\Theta) < 0$.

Given the highly nonlinear and implicit nature of the structural responses derived from the fully coupled time-domain simulations, this study employs a widely-used integral method for estimating failure probabilities based on the probabilistic fitting of performance functions (e.g., Li et al., 2024; Wang et al., 2024). Firstly, the peak structural responses at the mudline (i.e., y_{cal} and θ_{cal}) are extracted from respective time histories. Optimal probability distributions are fitted with a focus on the tail behavior of these extreme responses. Then, the probability of failure, P_f , can be computed by:

$$P_f = 1 - F_g(g_{lim}) \quad (15)$$

where g denotes the limit state function of y or θ ; $F_g(\bullet)$ is the fitted cumulative distribution function of g ; and g_{lim} is the allowable threshold. To maintain consistency with engineering practice, the reliability index, β , is also calculated by:

$$\beta = -\Phi^{-1}(P_f) \quad (16)$$

where $\Phi(\bullet)$ is the cumulative distribution function of standard normal distribution. This method provides a unified quantitative metric to

Table 3
Probabilistic models related to multi-source uncertainties.

Parameter		Distribution	Mean	Coefficient of variation	Reference
Wind speed		Normal	Table 2	0.05	Okpokparoro and Sriramula (2021)
Wave height		Normal	Table 2	0.05	
Wave period		Normal	Table 2	0.05	
Current speed		Normal	Table 2	0.05	
Internal friction angle	Soil layer 1	Lognormal	33.0°	0.08	Han et al. (2025)
	Soil layer 2		35.0°	0.08	
	Soil layer 3		38.5°	0.08	
	Soil layer 1		10 kN/m ³	0.05	
Effective unit weight	Soil layer 2	Normal	10 kN/m ³	0.05	—
	Soil layer 3		10 kN/m ³	0.05	
	Soil layer 1		5 m	0.1	
Layer thickness	Soil layer 2	Dependent ^a	9 m	0.1	
	Soil layer 3		22 m ^a	—	

^a Note: The thickness of soil layer 3 is a dependent variable dynamically calculated to ensure the total embedded pile length remains constant at 36 m.

Table 4
Comparison of the first two natural frequencies of the support structure.

Mode type	Frequency (Hz)	OC3 report (Hz)	Relative error (%)
1st tower fore-aft	0.240	0.246	−2.4
1st tower side-to-side	0.244	0.248	−1.6
2nd tower fore-aft	1.535	1.533	0.13
2nd tower side-to-side	1.544	1.546	−0.13

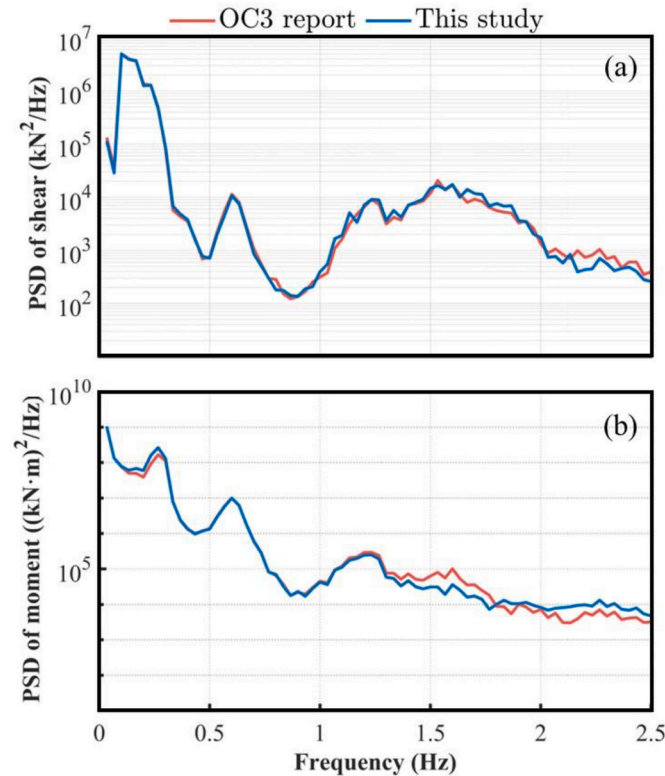


Fig. 5. Validation of power spectral density (PSD) of the foundation at the mudline: (a) shear force; (b) bending moment.

compare the structural safety margins between the coupled and decoupled methods under multi-source uncertainties within an acceptable computational cost.

3. Decoupled reliability analysis framework

To quantify the inherent errors in the industry-standard split-design

practice, this section establishes a simplified decoupled analysis framework to simulate the interaction between turbine manufacturers and foundation designers. Fig. 3 compares the mechanical transmission and uncertainty propagation of coupled and decoupled methods. The decoupled method is executed through a two-phase procedure. In Phase I, an independent aero-hydro-servo-elastic analysis of the upper structure supported by a rigid foundation is conducted to extract interface loads, in which only uncertainties in winds and sea states are considered. In Phase II, the pre-calculated time histories of interface loads are applied to a flexible foundation substructure to incorporate the effects of random soils. This two-phase framework focuses on reproducing the dynamic behavior of the physical system after decomposition into sub-systems, thus maintaining consistent multi-source uncertainty quantification with the fully coupled model as described in Section 2.2.

3.1. Phase I: upper structure dynamic analysis and load extraction

In Phase I, the upper structure is decoupled from the foundation and analyzed independently. The random wind and wave loads are generated via the AeroDyn and HydroDyn modules in OpenFAST, respectively, while structural dynamics are solved using the SubDyn module. The complete time history of the six-degree-of-freedom load at the foundation interface (mudline), $\mathbf{F}_{interface}(t)$, is extracted as:

$$\mathbf{F}_{interface}(t) = [H_x(t), H_y(t), H_z(t), M_x(t), M_y(t), M_z(t)]^T \quad (17)$$

where H_x , H_y , and H_z are the lateral shear forces at the mudline along the x , y , and z axes, respectively; and M_x , M_y , and M_z are the overturning bending moments at the mudline about the x , y , and z axes, respectively.

Different from the peak or envelope loads used in many simplified methods, the complete load history can preserve the information of phase and frequency and have a better consistency with the fully coupled method. It is crucial to highlight that, although the stochasticity of aerodynamic and hydrodynamic loads is preserved, the assumption of a rigid foundation prevents the real-time feedback of soil stiffness to the upper structure. Consequently, the suppression of the tower's first-mode vibration by aerodynamic damping is evaluated based on a fixed natural frequency, ignoring the frequency migration effect caused by foundation flexibility.

3.2. Phase II: foundation dynamic analysis and reliability assessment

In Phase II, the monopile foundation is treated as an independent substructure using the identical soil-pile interaction modeling with geotechnical uncertainties introduced via the random stiffness matrix. The interface load $\mathbf{F}_{interface}$ extracted in Phase I is applied as external excitations at the pile head. The governing equation of motion for this decoupled foundation can be expressed as:

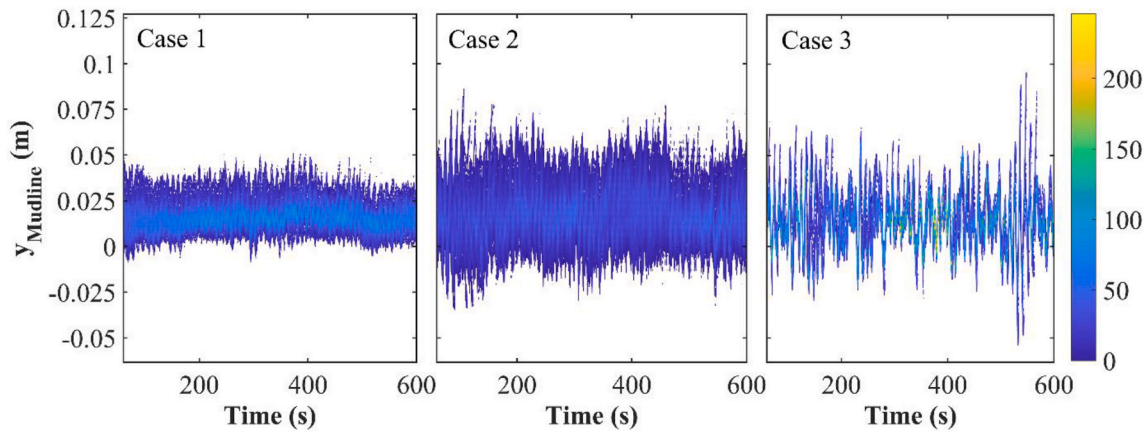


Fig. 6. Probability density contours of the pile-head lateral displacement.

$$\mathbf{M}_f \ddot{\mathbf{u}}_f(t) + \mathbf{C}_f \dot{\mathbf{u}}_f(t) + \mathbf{K}_f \mathbf{u}_f(t) = \mathbf{F}_{interface}(t) \quad (18)$$

where $\mathbf{M}_f$, $\mathbf{C}_f$, and $\mathbf{K}_f$ are the mass, damping, and stochastic stiffness matrices, respectively, of the foundation; and $\mathbf{u}_f$, $\dot{\mathbf{u}}_f$, and $\ddot{\mathbf{u}}_f$ are the nodal displacement, velocity, and acceleration vectors, respectively. The governing equation can be solved in the stand-alone SubDyn module to obtain the lateral displacement and rotation angle of the pile head.

This one-way coupling method severs the closed-loop dynamic feedback of the holistic system. In a fully coupled model, instantaneous variations in soil stiffness or extreme wave loads induce tower-top displacements, which in turn dynamically alter the aerodynamic angle of attack and rotor thrust, forming a continuous global dynamic feedback loop. In the decoupled model, however, $\mathbf{F}_{interface}$ is pre-calculated and fixed, rendering the upper structure loads entirely unresponsive to dynamic state changes on the foundation side. This truncation effect of uncertainty propagation will be quantified later in Section 4.

Once the lateral displacement and rotation angle of the pile head are obtained, the reliability indices regarding serviceability and ultimate limit states can be obtained using the same method described in Section 2.3. This alignment ensures that any observed discrepancies in the final reliability metrics are solely attributable to the physical modeling differences (i.e., coupled feedback vs. decoupled truncation) rather than inconsistencies in the probabilistic assessment method.

4. Illustrative example

This section compares the proposed coupled and decoupled reliability analysis frameworks using a widely-adopted NREL 5 MW reference OWT supported by an OC3 monopile foundation (Jonkman et al., 2009; Jonkman and Musial, 2010). Since this reference turbine is designed based on industry trends and data from other reference turbines, the results presented in this study should be interpreted as indicative references for OWT foundation reliability analysis rather than as a basis for normative calibration.

4.1. OWT model

Table 1 summarizes the key parameters of the reference turbine (Jonkman et al., 2009). Three-dimensional turbulent wind fields are generated by TurbSim and input to AeroDyn for aerodynamic analysis, with a rectangular grid covering the rotor swept area (lateral and vertical extents of 178×178 m and a grid resolution of 31×31 points). Regarding the sea state, irregular wave kinematics are generated using the JONSWAP spectrum combined with Stokes second-order wave theory in SeaState, coupled with a steady subsurface current profile, and then transmitted to HydroDyn for hydrodynamic analysis. The geotechnical environment is based on the standard OC3 reference site

(Jonkman and Musial, 2010), as shown in Fig. 4. The monopile, modeled by SubDyn, is embedded in a three-layer sandy seabed, with each stratum exhibiting distinct mechanical properties and thicknesses. For dynamic analysis, the simulation is run for 660 s, with the first 60 s discarded to eliminate initial transient effects and the subsequent 600 s used for time-domain analysis.

4.2. Load cases

The selection of load cases follows Morató et al. (2017) and Jonkman and Musial (2010), which identified the crucial subset of ultimate limit state load cases defined in IEC 61400-3-1 (International Electrotechnical Commission, 2019) and designed load cases for OC3 project, respectively. Three representative load cases (Table 2) are considered:

- (1) Case 1 (LC 5.2): The focus is the effects of operational loads and aerodynamic damping under severe sea state. The mean wind speed corresponds to the turbine's rated wind speed of 11.4 m/s. Case 1 represents a relatively conservative normal operation scenario.
- (2) Case 2 (DLC 1.3): The focus is the structural response under extreme turbulence. The mean wind speed of 14 m/s exceeds the rated speed but remains below the cut-out speed of 25 m/s. Case 2 represents an extreme turbulence scenario during power production.
- (3) Case 3 (DLC 6.2a): The focus is the dominance of hydrodynamic loads in the absence of aerodynamic damping. Case 3 represents an extreme scenario where the turbine is parked (idling) due to a loss of the electrical network connection.

4.3. Uncertainty modeling

Table 3 summarizes the probabilistic models for multi-source uncertainty characterization. With respect to the aerodynamic and hydrodynamic uncertainties, the four parameters, i.e., wind speed (U_w), wave height (H_s), wave period (T_p), and current speed (U_c), are modeled as normal random variables with mean values defined in corresponding load cases (Table 2) and a consistent coefficient of variation of 0.05 (Okpokparoro and Sriramula, 2021). Herein, the assumption of statistically independent normal distributions for H_s and T_p is a case-study-specific simplification, since the primary objective of this study is to compare reliability differences between coupled and decoupled methods, rather than to deliver a precise site-specific reliability calibration.

Regarding the geotechnical uncertainties, the internal friction angles and effective unit weights of the three soil layers are modeled as lognormal and normal random variables, respectively, with coefficients

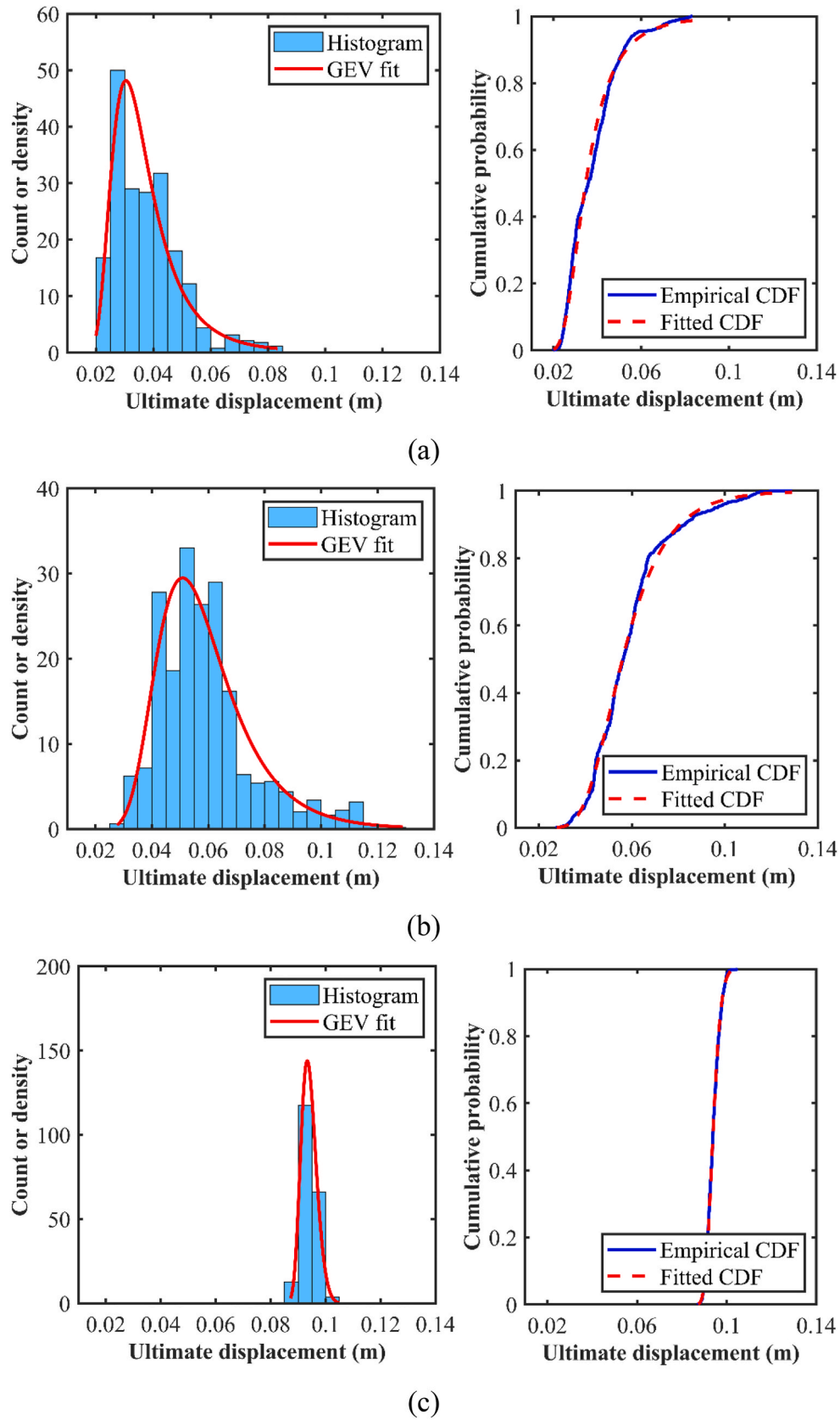


Fig. 7. Probability distributions of ultimate displacement: (a) Case 1; (b) Case 2; (c) Case 3.

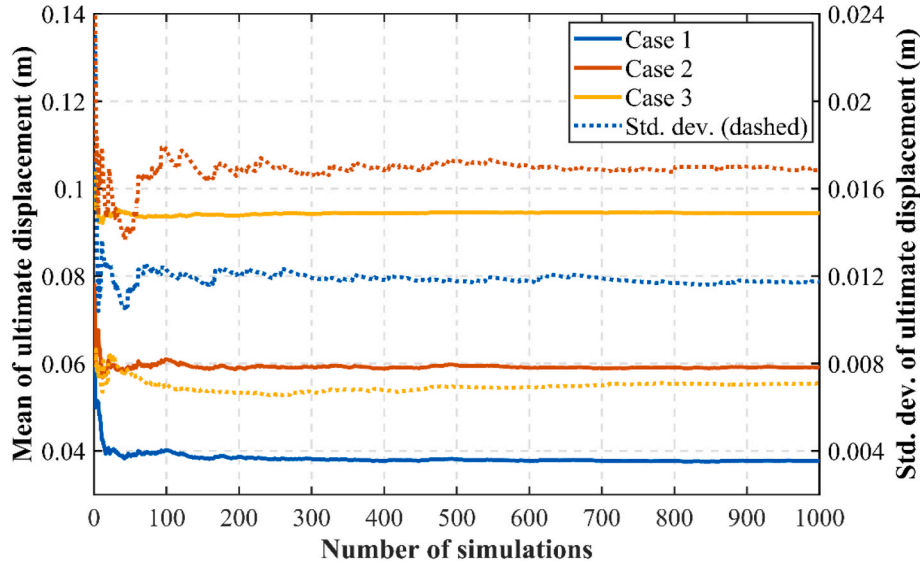


Fig. 8. Convergence analysis of mean and standard deviation of ultimate displacement.

of variation of 0.08 and 0.05, respectively, following Han et al. (2025). The thicknesses of soil layers 1 and 2 are modeled as normal random variables with coefficient of variation of 0.1, while the thickness of soil layer 3 is a dependent variable dynamically calculated to ensure the total embedded pile length remains constant at 36 m (i.e., deducting the randomly sampled thicknesses of layers 1 and 2 from 36 m). Compared with the soil profile in an actual offshore site that typically exhibits very high spatial discreteness, the coefficients of variation adopted in this study are relatively conservative.

4.4. Integrated reliability analysis results

4.4.1. Deterministic soil-pile interaction analysis

The monopile supporting the OWT has an outer diameter of 6.04 m, a wall thickness of 0.067 m, and a total length of 68.6 m, with embedded length being 36 m. As described in Section 2.1.2, this study employs the coupled springs model to represent the monopile as a flexible foundation. The resulting deterministic pile-head stiffness matrix (i.e., Eq. (1)) is obtained as:

$$\mathbf{K}_{soil} = \begin{bmatrix} 1.83 & 0 & 0 & 0 & -14.9 & 0 \\ 0 & 1.83 & 0 & 14.9 & 0 & 0 \\ 0 & 0 & 9.57 & 0 & 0 & 0 \\ 0 & 14.9 & 0 & 187 & 0 & 0 \\ -14.9 & 0 & 0 & 0 & 187 & 0 \\ 0 & 0 & 0 & 0 & 0 & 163 \end{bmatrix} \times 10^9 \text{ N} / \text{m} \quad (19)$$

This stiffness matrix is continuously re-evaluated and dynamically updated in each random realization to propagate the geotechnical uncertainties defined in Table 3.

It is important to verify the accuracy of the numerical model for the OWT support system prior to conducting stochastic analysis. The OC3 project (Jonkman and Musial, 2010) provides eigen-analysis results for the full system with coupled springs boundaries in a parked state and the power spectral densities of the foundation under different load cases. They can be used to validate the model established in this study. The validation comprises two steps: (1) In a parked state, SubDyn is executed first with the soil-pile interaction input to generate the mass matrix MBBt and stiffness matrix KBBt of the platform, then these matrices are incorporated into BModes to compute mode shapes and natural frequencies of the structure accounting for foundation flexibility; (2) Under the environmental condition of Case 1, the parameter files characterizing the wind, wave, current, and flexible foundation are integrated into OpenFAST to execute a fully coupled time-domain simulation and

extract the dynamic responses of the foundation at the mudline. Table 4 compares the first two natural frequencies of the support structure and Fig. 5 compares the power spectral densities of shear force and bending moment of the foundation at the mudline. The results obtained in this study agree well with those in the OC3 report, confirming the accuracy of the numerical model.

4.4.2. Dynamic responses under uncertainties

To understand the structural behavior under multi-source uncertainties, the overall dynamic responses of the foundation are first examined in the entire time domain (i.e., 600 s). Fig. 6 visualizes the probability density contours of the time histories of the pile-head lateral displacement across all 1000 stochastic simulations for the three load cases. Case 1 leads to the minimum displacement with a moderate variability; Case 2 leads to a moderate displacement with the highest variability; and Case 3 leads to the maximum displacement but with the lowest variability. This is expected as the turbine is parked in the extreme scenario Case 3.

Since extreme responses are the primary concern for structural safety, the peak values from the time histories in Fig. 6 are extracted to obtain the ultimate displacements of the foundation for statistical analysis. Fig. 7 displays the histograms of the ultimate displacement alongside their best fitted probability density curves. Five candidate probability distributions are considered, including normal, lognormal, Weibull, generalized extreme value (GEV), and generalized Pareto distributions. Based on the Akaike information criterion, the generalized extreme value distribution is identified as the optimal model in all the cases for describing the ultimate displacement. Again, it is found that Case 3 results in much higher ultimate displacement but with much smaller variability.

It should be noted that these 1000 random simulations are utilized specifically to propagate multi-source uncertainties and construct the probability distributions of ultimate displacement, rather than to compute failure probabilities using direct Monte Carlo simulation. Convergence analysis (Fig. 8) confirms that both the mean and standard deviation of the ultimate displacement converge well before reaching 1000 simulations, validating the statistical sufficiency of the sample size in this study.

4.4.3. Reliability analysis under the coupled framework

Fig. 9(a) and Fig. 9(b) present the reliability indices (β) calculated based on extreme value distributions for the lateral displacement and rotation limit states, respectively. To explore the impact of geotechnical

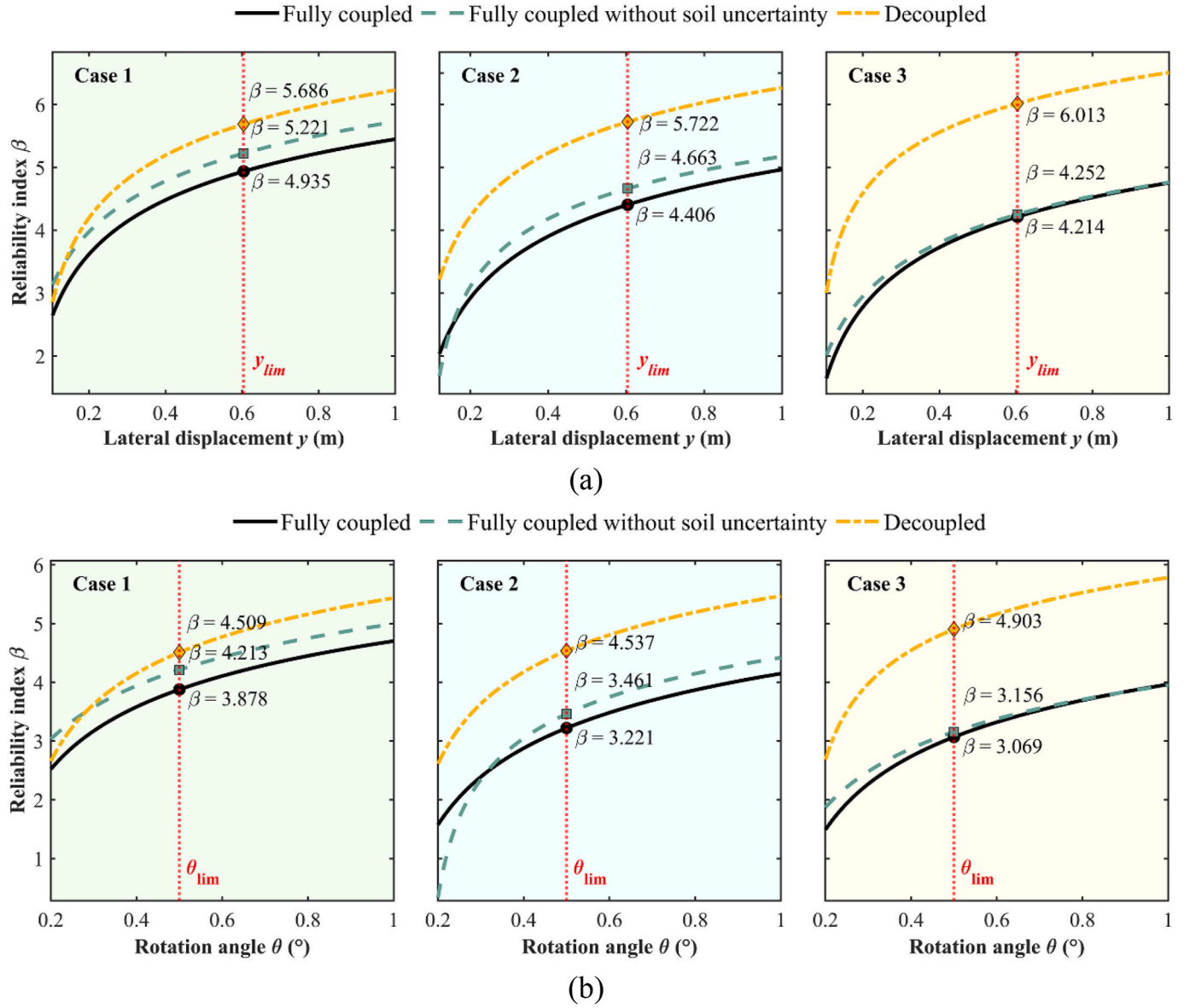


Fig. 9. Comparison of coupled and decoupled reliability analysis results: (a) lateral displacement limit state; (b) rotation limit state.

uncertainties, these figures compare the coupled baseline incorporating all multi-source uncertainties (black solid lines) against a hypothetical coupled scenario that explicitly ignores soil variability (green dashed lines). Given the allowable limits of $y_{lim} = 0.1D = 0.604$ m and $\theta_{lim} = 0.5^\circ$, the reliability indices considering soil variability for the three load cases are 4.935, 4.406, and 4.214, respectively, for lateral displacement limit state and 3.878, 3.221, and 3.069, respectively, for rotation limit state. Ignoring geotechnical uncertainties will result in a modest overestimation of reliability index by 1%–9% across the three load cases. As expected, geotechnical uncertainties exert a more pronounced impact under operational load cases. As load cases become increasingly extreme, the influence of load uncertainties considerably surpasses that stemming from geotechnical uncertainties. However, considering that the difference will widen as realistic geotechnical uncertainties usually reach a significantly higher level (e.g., coefficient of variation > 0.3), it is still necessary to incorporate geotechnical uncertainties into OWT reliability assessment through a fully coupled framework.

4.5. Comparison of coupled and decoupled methods

To investigate the inherent modeling inadequacies in the traditional split-design approach, this section makes a comprehensive comparison between the fully coupled method and the simplified decoupled method proposed in Section 3. To achieve fair comparison, the 1000 random realizations used in the coupled method are re-evaluated.

4.5.1. Interface loads from the upper structure

The decoupled method assumes a rigid foundation in Phase I, yielding a first-order natural frequency of the tower of approximately 0.275 Hz, which is about 15% higher than the flexible foundation used in the coupled method (i.e., 0.240 Hz as shown in Table 4). Through dynamic analysis of the upper structure, the decoupled method calculates the interface forces and moments applied to the pile head. Fig. 10 extracts the peak values of the interface loads at the mudline from their time histories and compares the results from both decoupled and coupled methods. The peak forces and moments of the decoupled method are significantly larger than those of the coupled method. The linear correlations between the two methods are particularly high for

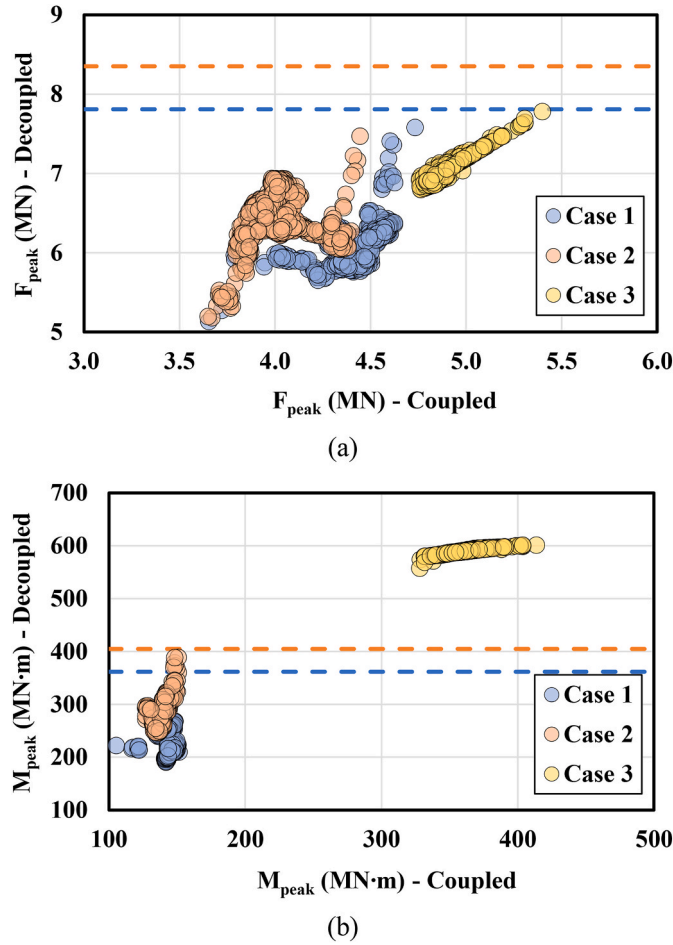


Fig. 10. Comparison of interface loads: (a) peak force; (b) peak moment.

Case 3, i.e., 0.96 for force and 0.95 for moment. By comparison, Fig. 10 also plots the equivalent ultimate loads calculated according to the design code IEC 61400-3-1 (see Appendix A) by the dashed lines. They are always greater than those obtained using the proposed decoupled method in all cases. The equivalent ultimate loads in Case 3 are not provided since they significantly exceed the range, more than twice those obtained using the proposed decoupled method. The normative

design codes, typically employed in preliminary design phases, utilize efficient but low-fidelity decoupled methods with partial factors to broadly cover all uncertainties. Consequently, they tend to be highly conservative, a trend visibly confirmed in Case 1 and Case 2 of this study. Case 3 represents an extreme environmental condition where the turbine is parked, and the control system is active to mitigate adverse structural responses. Simplified normative calculations typically fail to account for such active control interventions, leading to great biases in Case 3.

4.5.2. Key responses of the foundation substructure

Based on the interface loads obtained in the previous step, the maximum lateral displacement and rotation angle of the foundation substructure at the mudline given the same geotechnical uncertainties can be calculated using the proposed decoupled method. Fig. 11 compares the box plots of the structural responses obtained from the decoupled and coupled methods. For all the three cases, the response variation patterns predicted by the decoupled method agree with those of the coupled method, which indicates that the decoupled method can capture the mechanical transmission and uncertainty propagation to a certain degree. It is also evident that the decoupled method generally predicts larger structural responses (i.e., higher mean values), yet underestimates the associated variabilities (i.e., narrower value ranges). The former is attributed to the simplification of rigid foundation in Phase I, and the latter is due to the truncated uncertainty propagation in the decoupled method. To provide a quantitative benchmark of normative conservatism, the deterministic design code (IEC 61400-3-1) yields a maximum lateral displacement of 0.142 m and a rotation angle of 0.291° for Case 1 and $[0.168 \text{ m}, 0.345^\circ]$ for Case 2. These values significantly exceed the upper bounds of the probabilistic responses from both models presented in Fig. 11, which further confirms the substantial safety redundancy embedded in current split-design specifications.

4.5.3. Reliability assessment and comparison

Although the decoupled method overestimates structural response magnitudes, its implication for overall safety requires deeper probabilistic scrutiny. Fig. 9(a) and Fig. 9(b) superimpose the reliability indices (β) obtained from the decoupled method (orange dashed-dotted lines) onto the coupled baselines. Contrary to the expectation by the conservative (i.e., larger) response magnitudes, the reliability indices computed via the decoupled model are consistently higher than those obtained from the coupled benchmark across all load cases, and the

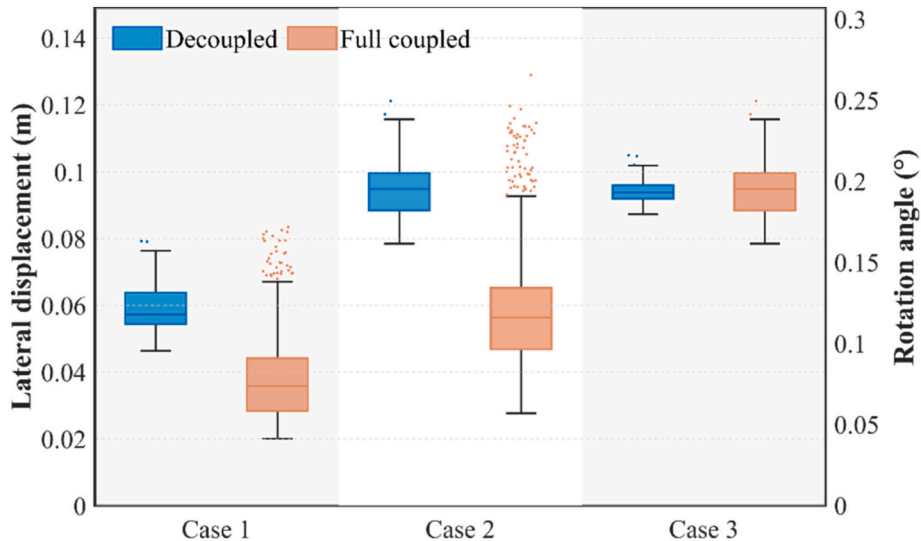


Fig. 11. Comparison of structural responses of the foundation.

discrepancy expands as the load case becomes increasingly extreme. Specifically, β values in three cases are increased by 15%, 30%, and 43%, respectively, for lateral displacement limit state, and 16%, 41%, and 60%, respectively, for rotation limit state. This implies that the split-design model overestimates the reliability of OWT foundation, making it non-conservative in a probabilistic sense. This paradox is attributed to a significant reduction in response variability, as observed in Fig. 11.

Notably, the decoupled method relies on a one-way sequential load transfer, which inherently truncates the continuous, bidirectional dynamic interaction between the upper structure and the foundation. By severing this physical continuity, the decoupled method filters out the complex aero-hydro-geotech feedback loops that amplify system response variability under environmental uncertainties. Consequently, a substantial portion of the multi-source uncertainties is lost during the interface transmission. It is this truncation of uncertainty propagation that inflates the reliability index and engenders a perilous “false sense of security”. These findings demonstrate a critical, yet frequently overlooked vulnerability in traditional split-design practice and highlight the necessity of fully coupled probabilistic frameworks for reliability-based design of future offshore wind developments.

5. Summary and conclusions

This study establishes a fully coupled aero-hydro-geotech reliability analysis framework to quantify how multi-source uncertainties affect the safety of OWTs. It integrates soil-pile interaction modeling, time-domain dynamic simulation, and multi-source uncertainty modeling into the OpenFAST platform. A simplified decoupled method is also developed to benchmark the industrial split-design practice. These methods are validated and compared through an NREL 5 MW reference OWT under three IEC design load cases. The main conclusions drawn from this study are as follows:

- (1) Geotechnical uncertainties from soil parameters and stratigraphic conditions can be properly considered through a variable pile-head stiffness matrix in soil-pile interaction modeling, which alters the OWT's natural frequency and dynamic response. Since ignoring geotechnical uncertainties will result in a modest

overestimation of reliability, particularly under operational load cases, it is necessary to incorporate geotechnical uncertainties into OWT reliability assessment through a fully coupled framework.

- (2) The proposed decoupled method can capture the mechanical transmission and uncertainty propagation in the coupled method to a certain degree. It outperforms the current split-design specifications that utilize efficient but low-fidelity decoupled methods with partial factors to broadly cover all uncertainties and provide unnecessarily excessive design redundancy.
- (3) A counter-intuitive “false sense of security” emerges from the decoupled method. Although it yields higher (conservative) mean responses due to the simplification of rigid foundation, the truncation of uncertainty propagation during decoupling significantly reduces response variabilities, resulting in non-conservatively high reliability indices, with an overestimation ranging from 15% to 60%. The coupled method can provide a robust tool for design code calibration under multi-source uncertainty propagation to mitigate the excessive design redundancy issue.

CRedit authorship contribution statement

Heng Zhou: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis. **Te Xiao:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Lulu Zhang:** Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Environmental load calculations (IEC 61400-3-1)

To benchmark the proposed simplified decoupled model against the standard industrial practice, this appendix provides the calculations of ultimate environmental loads in IEC 61400-3-1 (International Electrotechnical Commission, 2019) in terms of horizontal thrust and overturning moment.

Based on the extreme turbulence model of wind, the ultimate aerodynamic thrust F_{wind} and overturning moment M_{wind} at the mudline can be evaluated as:

$$F_{wind} = \frac{1}{2} \rho_a A_R C_T (U_w + U_{ETM})^2 \quad (A.1)$$

$$M_{wind} = F_{wind} \cdot (S + z_{hub}) \quad (A.2)$$

where ρ_a is the air density; A_R is the rotor-swept area; C_T is the thrust coefficient; U_w is the mean wind speed at the hub height; U_{ETM} is the extreme turbulent wind speed; S is the water depth; and z_{hub} is the hub height above sea level.

Given that many OWTs are located in shallow waters, hydrodynamic forces can be calculated using the Morison's equation. The total horizontal hydrodynamic force per unit length, f_H , is expressed as:

$$f_H = \frac{1}{2} C_D \rho D (U_c + U_x) |U_c + U_x| + C_M \rho \frac{\pi D^2}{4} \frac{\partial U_x}{\partial t} \quad (A.3)$$

where C_D and C_M are the drag and inertia coefficients, respectively; ρ is the seawater density; D is the pile diameter; U_c is the current speed; and U_x is the water particle velocity in the horizontal plane. The terms U_x and $\partial U_x / \partial t$ can be determined using the Stokes second-order wave theory. Eventually, the total shear force F_{wave} and overturning moment M_{wave} acting on the pile at the mudline can be obtained by integrating the distributed $f_H(z, t)$ from the seabed (mudline at $z = -d$) up to the instantaneous wave surface elevation $\eta(t)$ as:

$$F_{\text{wave}}(t) = \int_{-d}^{\eta(t)} f_H(z, t) dz \quad (\text{A.4})$$

$$M_{\text{wave}}(t) = \int_{-d}^{\eta(t)} f_H(z, t) \cdot (z + d) dz \quad (\text{A.5})$$

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